Dear examiner,

please find the criteria marks as well as teacher's comments below.

Student: dv621

A/4	B/4	C/3	D/3	E/6	Total
4	4	2	3	6	19

Criterion A: Presentation
A well-written, logically developed, and well-structured IA. The student successfully introduces the mathematics required to solve a real-life problem of his choice, using a number of comprehensive examples along the way. Graphs, tables, equations and diagrams are used effectively to aid understanding. The IA has an introduction with a clear and focused aim, which is met. A conclusion summarizes the findings. There are no irrelevant or repetitive calculations.

Criterion B: Mathematical communication
Appropriate and consistent mathematical language is used at all times. The mathematical notation is error-free. Unfamiliar key terms are defined and explained well. Formulas, a large number of diagrams, and a table are used. His rather narrative induction proof on pages 15 and 16 is original and set out logically.

Criterion C: Personal engagement
A very creative and high quality IA on a fascinating topic chosen by the student who really made this exploration his own. The solution on pages 6 and 7, the induction proof, and all diagrams are original. The IA contains lots of unfamiliar mathematics and the student has done a great job independently investigating these concepts (including the algorithm) and explaining them well so that a fellow student can comprehend.

Criterion D: Reflection
The student consistently reflects on his mathematical working - critical reflection is seen at any stage really. It helps the IA develop, move forward, and gives it a very nice flow. Implications of the results are discussed, possible extensions can be found in the conclusion. A number of insightful examples aid understanding.

Criterion E: Use of mathematics
All explanations of mathematical arguments are clear and logical. The student utilizes a number of relevant mathematical concepts: ratios, 2D and 3D geometry, combinations, angles between vectors, convex hulls, and mathematical induction. The student explores the problem from different perspectives; all mathematical working is error-free.

Can we mix it? An exploration into creating target colors from existing colors using convex hulls

Student code: hdv621

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Introduction:

As a person interested in art, I was always curious as to how the 3 primary colors, red, yellow, and blue could make all possible color combinations. Since I did not have all the possible colors, I wondered what colors I could make using my current paint collection. In order to make new colors using blends of pre-existing colors, one must consider the colors that are already in existence and check if it is possible to make a target color by mixing pre-existing colors in certain ratios. For simplicity, the quantity of available paint will not be considered, and it will be assumed that there is an infinite quantity of each type of paint.

The problem definition is as follows. Given n unique pre-existing colors that are each given in the ratio $r: y: b$, check if it is possible to make a target ratio of $r_{target}: y_{target}: b_{target}$ and in the case that it is possible, find the minimum number of pre-existing colors that have to be mixed. For simplicity, we will assume that the ratio is given in natural numbers and that $r, y, and\ b$ are given in coprime form (i.e. $r, y, and\ b$ do not have a common divisor that is greater than 1).

This investigation aims to come up with a solution to this problem for a general case and try the solution out on sample sets of pre-existing and target colors. In our case, we will have 8 pre-existing colors that are part of my current paint collection and check if 3 target colors can be made. This will be done in the real-life applications section. In order to do this, we will make some key observations that makes the problem easier to solve and afterwards, come up with a working solution. Finally, we will optimize the solution to come up with a more optimized and elegant solution. Furthermore, along the way we will prove and/or explain in detail all relevant statements and lemmas that are used to solve the problem.

Modelling the problem:

When we first look at a few ratios, it may be hard to see which colors can be made using those ratios. For example, given the ratios $(1: 2: 1), (3: 2: 1)$ it can be quite hard to see which ratios can be made using the two given ratios. For example, what is the ratio of the color that is made when 2 parts of the first ratio $(1: 2: 1)$ is mixed with 1 part of the second ratio $(3: 2: 1)$? Furthermore, this way of representing ratios has a problem of similar ratios looking quite different. For example, the two ratios $(1: 2: 3)$ and $(7: 14: 22)$ are almost similar ratios but when looking at the two, they look to be quite different.

In order to alleviate these 2 issues, we can map every ratio to a point in a 3D space using the following rule:

$$(r: y: b) = \left(\frac{r}{(r + y + b)}, \frac{y}{(r + y + b)}, \frac{b}{(r + y + b)}\right)$$

By mapping the ratios to a point in 3D space, we get some interesting properties.

Property 1: The sum of the x, y, z coordinate will always be 1:

$$\frac{r}{(r + y + b)} + \frac{y}{(r + y + b)} + \frac{b}{(r + y + b)} = \frac{(r + y + b)}{(r + y + b)} = 1$$

Property 2: Using property 1, we can see that when combining 2 points with coordinates (x_1, y_1, z_1) and (x_2, y_2, z_2) we can multiply the first point by q_1 and the second by q_2 such that q_1 and q_2 sum to 1. This will give a final point of $(x_1 \cdot q_1 + x_2 \cdot q_2, y_1 \cdot q_1 + y_2 \cdot q_2, z_1 \cdot q_1 + z_2 \cdot q_2)$.

Property 3: Any point made using property 2 will obey property 1. Therefore, we can say that any ratio that is mapped to a point will lie inside a $1x1x1$ cube in 3D space and every x-, y-, and z- coordinate will be in the range $[0,1]$.

By mapping every ratio to a point in 3D space, the 2 main issues that we previously had are solved. First, it is quite easy to know which points can be made by combining 2 different points by using the second property. Also, similar ratios will have similar x-, y-, and z-coordinates as the coordinates are representative of the percentage of the sum of all values $(r+y+b)$. For example, the ratios $(1:2:3)$ and $(7:14:22)$ will now evaluate to points $(\frac{1}{6},\frac{1}{3},\frac{1}{2})$ and $(\frac{7}{43},\frac{14}{43},\frac{22}{43})$ respectively. We can see that the 2 points are quite similar unlike the 2 ratios meaning that by using this point system instead of the ratio system, we are able to better represent the ratios in a way such that similar ratios have similar values.

We will now look at how combining points can result in new points being made. We will divide this section into 4 sections as follows. A) only using 1 point, B) only using 2 points, C) only using 3 points and finally D) using more than 3 points.

A) Only using 1 point:

This section is extremely simple, the only point that can be made using 1 point is the point itself because you cannot multiply the point by a number that is not equal to 1 as it will mean that this new point will not satisfy the property that the sum of the x-, y-, and z-coordinate is equal to 1. Also, in reality, if you were to only have 1 color of paint, the only color that you could make is the color that you already have.

B) Only using 2 points:

This case starts to become interesting. Let us first assume that the 2 points that we have are $P1=(x_1,y_1,z_1)$ and $P2=(x_2,y_2,z_2)$ where $P1$ and $P2$ are different points.

In mathematics, we learned that the midpoint of a line segment is equal to $(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2},\frac{z_1+z_2}{2})$. This is equivalent to $(x_1\cdot 0.5+x_2\cdot 0.5, y_1\cdot 0.5+y_2\cdot 0.5, z_1\cdot 0.5+z_2\cdot 0.5)$. Looking at this point, we can say that it is equal to $P1\cdot 0.5+P2\cdot 0.5$. Now we can replace the coefficients of the 2 points with a_1 and a_2 where $a_1+a_2=1$ such that the new point becomes $P1\cdot a_1+P2\cdot a_2$. We can see that as a_1 gets closer to 1, the new point that is created is closer to point $P1$. On the other hand, as a_2 gets closer to 1, the new point that is created is closer to point $P2$. By using different values of a_1

and a_2 it will always be possible to make any point that lies on the line segment between point $P1$ and $P2$. Whilst a formal proof is possible, we can think of this as follows. If $a_1 = a_2 = 0.5$ then the new point that is created will be the midpoint of line segment $P1P2$. Now, if we were to increase a_1, we would move farther away from the midpoint and closer to $P1$. Since a_1 can take on any real value in the range $[0,1]$, it must be possible to make any point between the point $P1$ and the midpoint. We can also swap a_1 with a_2 and see that by increasing a_2, the point that is created will become closer to $P2$.

Therefore, by combining 2 points, we can create any new point that lies on the line segment between the 2 points meaning that we can create any color that has a mapping that lies on the segment between the 2 points.

C) Only using 3 points:

This case is also interesting as the number of points that we can make are infinite. We can first assume that the 3 points $P1 = (x_1, y_1, z_1)$, $P2 = (x_2, y_2, z_2)$ and $P3 = (x_3, y_3, z_3)$ are all distinct and non-colinear (the 3 points do not lie on one line). In order to find out which points can be made using 3 points, we can repeat what we did when we only had 2 points twice. First, we pick 2 points (we can assume that this is $P1$ and $P2$). We now know that we can create any point that lies on the line segment between $P1$ and $P2$. We will call this point $P4$ from now on ($P4$ can be any point that lies on the line segment between $P1$ and $P2$). Now using $P3$ and $P4$, we can create any point that lies on the line segment between $P3$ and $P4$.

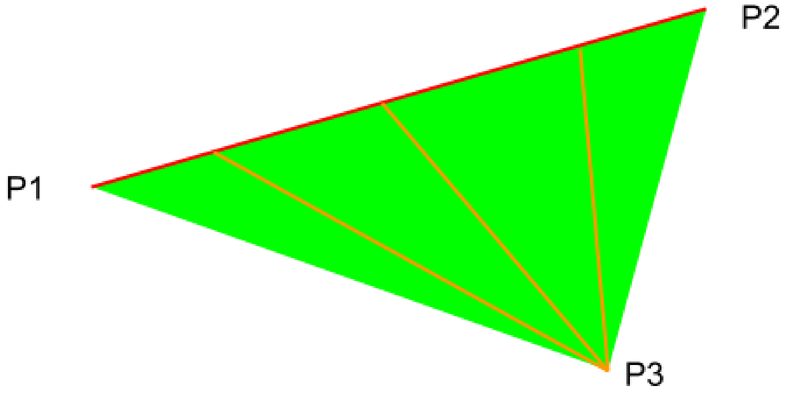

Figure 1: Possible target points using 3 points (made by myself using google drawings)

Figure 1 illustrates the points that can be made using 3 points. The red line going from $P1$ to $P2$ illustrates all the points that can be made using the 2 points which is defined as $P4$. The orange lines show some possible points that can be made using the newly created point $P4$ which lies on the red line and $P3$. The green triangle shows the points that can be made using the points $P1$, $P2$ and $P3$ using the method outlined above.

D) Using more than 3 points:

When using more than 3 points, the idea of creating a plane in 3D space starts to break down. For example, if we imagine that we have 4 points that are in the shape of a tetrahedron, it would be impossible to make a plane that encompasses all 4 points. Therefore, using more than 3 points does not yield any new points that can be made which were previously impossible to make either using 1, 2 or 3 points. Therefore, this case does not have to be considered.

A solution to the problem using 3D geometry:

Looking at the 4 cases A, B, C, and D, we can observe that if a target point can be made, it can be made either using 1,2 or 3 points. We will now explore how we can check if a particular point can be made using 1,2 or 3 points.

Checking whether or not a point can be made using 1 point is quite easy. We can go through all the pre-existing points and check if the target point is equal to any of the pre-existing points. If there is a point that is equal to the target point, the target point can be made using only 1 point. If not, the target point cannot be made using only 1 point.

Checking whether or not a point can be made using 2 points is a bit more complicated. We can go through all $\binom{n}{2}$ ways of choosing 2 points out of the n points. We can then create line segments using the 2 points and afterwards, check if the target point lies on any of these segments. In order to check if a point lies on the line segment, we can first find the direction vector of the line segment. Afterwards, we can check if the distance from the target point to one of the 2 original points is a multiple of the direction vector. If it is not, the point does not lie on the line segment. If it is, we must further check if the x coordinate of the target point is between the two x-values and the y coordinate of the target point is between the two y-values.

Checking whether or not a point can be made using 3 points starts to become quite complicated. One way of doing it is outlined below.

We can go through all $\binom{n}{3}$ ways of choosing 3 points out of the n points. These 3 points will now form the vertices of the triangle. It is worth noting that these 3 points must not be colinear as this will make it impossible to create a triangle using the 3 points.

After choosing the 3 points, we can first create 3 vectors as shown in figure 2. These vectors will be labeled as $\overrightarrow{v1}, \overrightarrow{v2}, \overrightarrow{v3}$. It is important to make these vectors as direction vectors from the target point to the vertices of the triangle. Afterwards, we can find the clockwise angle between the direction vectors for all 3 possible pairs of direction vectors ($\{\overrightarrow{v1}, \overrightarrow{v2}\}, \{\overrightarrow{v2}, \overrightarrow{v3}\}, \{\overrightarrow{v1}, \overrightarrow{v3}\}$). These can be found using:

$$angle(\vec{v}, \vec{w}) = \arccos\left(\frac{v_x \cdot w_x + v_y \cdot w_y + v_z \cdot w_z}{|v| \cdot |w|}\right)$$

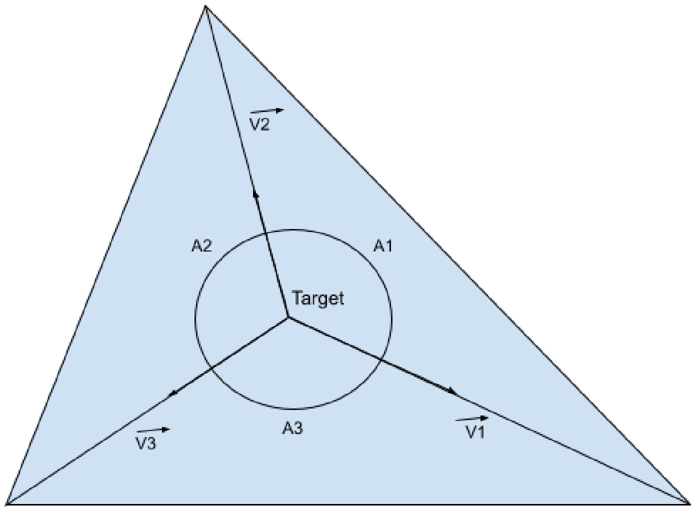

Figure 2: A point inside a triangle (made by myself using google drawings)

From figure 2, it is possible to see that if a point is inside the triangle, all the angles must be smaller than $180°$. Therefore, by checking if all the angles are smaller than $180°$, we can check whether or not a particular point is inside that triangle.

We have now come up with a solution that solves the original problem. We can first check if the target point can be made using 1 point and afterwards, 2 points and finally 3 points. If it is not possible to make the target point using any of these methods, we can conclude that it is impossible to make the target point. It should be mentioned that this solution was completely original and was not inspired by any other text.

Reduction to 2D geometry:

We have now found a way to solve the problem using $3D$ geometry. However, the solution is quite complicated and takes a lot of time as we would have to in the worst case, try all 3 cases to find out that the answer does not work. If the number of given points is rather large (e.g. 10000), the number of possible triangles would be around the order of 10^{11} which would take an extremely long time to check if the target point lies on any of those triangles. Therefore, we should come up with a solution that is a lot faster.

One important observation to make when reducing the problem into a $2D$ geometry problem is that we only have to consider the x- and y-coordinate. This stems from the fact that the sum of the x-, y- and z-coordinate of a point is equal to 1. This property is explained below.

Lemma: If the x- and y-coordinate of a point is equal to the x- and y-coordinate of the target point, the 2 points must be equal.

We will first call the point that we have $P1 = (x_1, y_1, z_1)$ and the target point $P2 = (x_2, y_2, z_2)$. We will also assume that $x_1 = x_2$ and $y_1 = y_2$.

We can substitute $x_1 = x_2$ and $y_1 = y_2$ giving $P1 = (x_2, y_2, z_1)$. We also know that $x_2 + y_2 + z_1$ and $x_2 + y_2 + z_2$ are equal to 1. Setting both equations equal to each other yields:

$$x_2 + y_2 + z_1 = x_2 + y_2 + z_2$$

$$\rightarrow z_1 = z_2$$

Therefore, if the x- and y-coordinates of a point are equal to the x- and y-coordinate of the target point, the 2 points must be equal.

The observation that we made in the $3D$ geometry solution, which was that if a point can be made, it can be made by combining 1, 2 or 3 points also holds in the $2D$ case. This is because there is no difference between the $3D$ and $2D$ solution except that in the $2D$ solution, we only consider the x and y coordinates instead of considering all the coordinates. It should be mentioned that the solution in $2D$ geometry was heavily inspired from the editorial of task mixtures which was a problem in the 2020 Baltic Olympiad in Informatics.

Convex hulls:

A convex hull is a convex polygon (all internal angles are smaller than 180°) such that it encloses all points and also has the minimum sum of edge lengths. Figure 4 shows the convex hull of the points shown in figure 3.

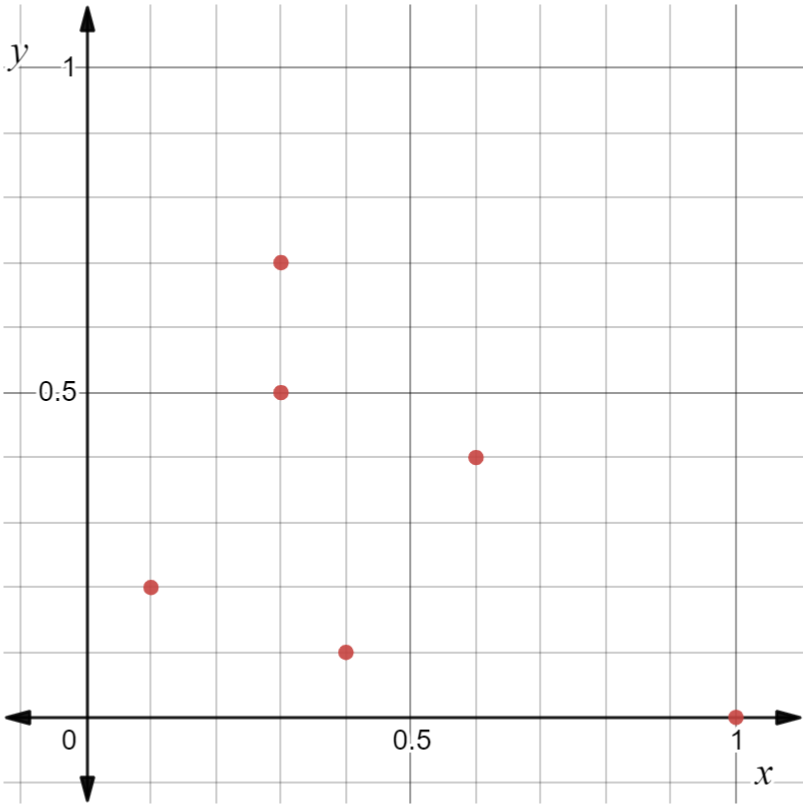

Figure 3: A cloud of points (made by myself using desmos)

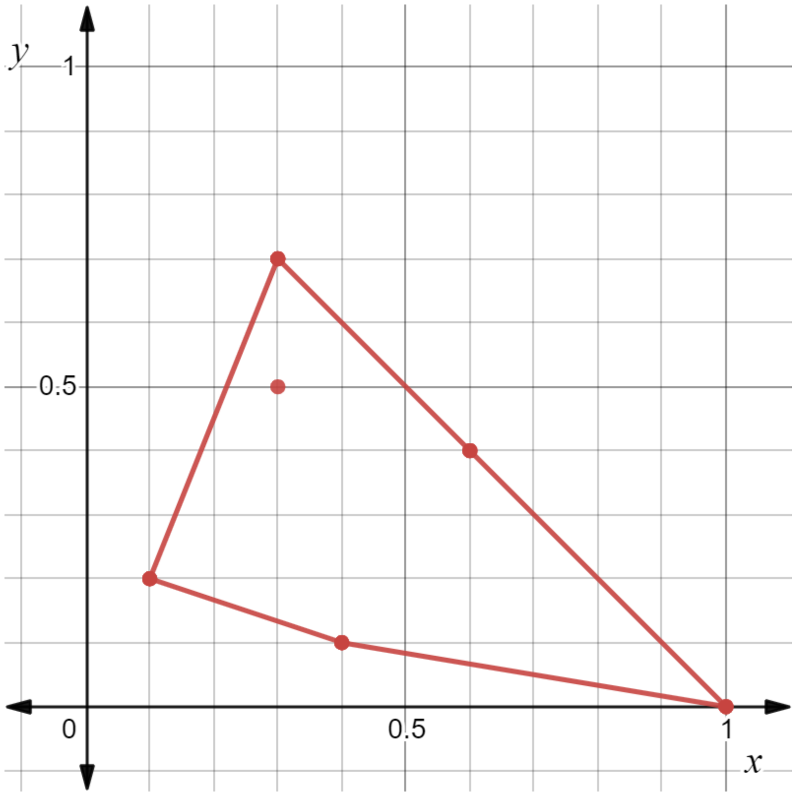

Figure 4: Example of a convex hull (made by myself using desmos)

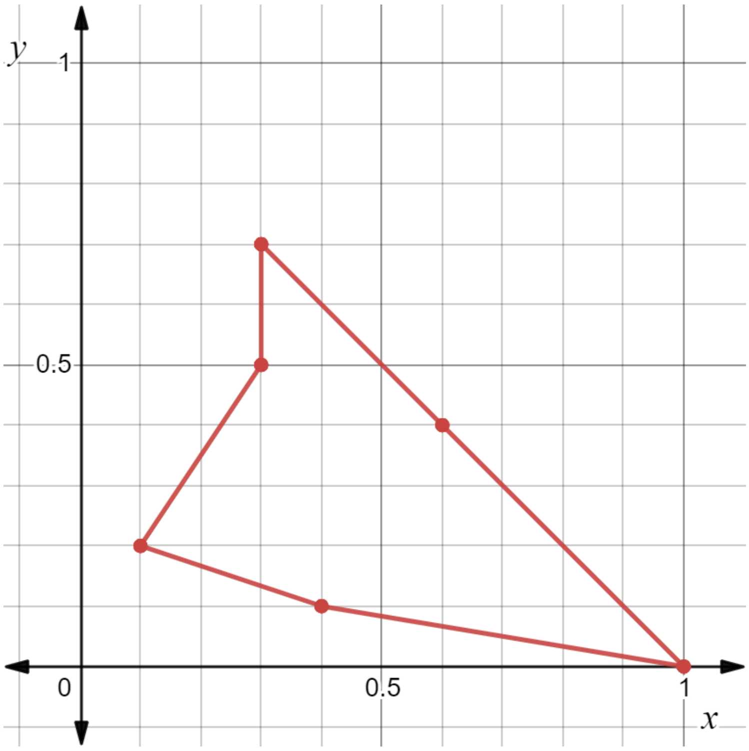

Figure 5: A polygon that is not a convex hull (made by myself using desmos)

Figure 5 is not a convex hull of the points shown in figure 3 because the internal angle at point $(0.3,0.5)$ is greater than $180°$ and therefore, does not constitute a convex polygon.

A convex hull is useful when solving the problem as if the target point is inside of the convex hull made using points mapped from pre-existing paint ratios, the target ratio can be made by blending 3 pre-existing colors. In other words, checking if a point is inside a convex hull takes place instead of checking whether or not a point is on a triangle created using 3 points.

Generating convex hulls:

There are many different algorithms that are capable of generating convex hulls all having their pros and cons. In this investigation, the gift-wrapping algorithm will be used due to its simplicity and ease of understanding. The algorithm is described below.

We must first find the leftmost point. This can be done by sorting the points by increasing x coordinate and in the case that there is a tie, by increasing y-coordinate. The leftmost point must be included in the convex hull as there is no way of enclosing that point in the convex hull other than by using the point itself. The rest of the convex hull can be found using the method below:

First, find the bearing between the last point that has been inserted into the convex hull (in the first iteration, the leftmost point) and every other point (except for the last point that has been inserted into the convex hull). Out of these find the one with the smallest angle. In the case that this point is a point that is already in the convex hull, terminate the algorithm or if it is not already in the convex hull, continue the algorithm by appending the newly found point into the convex hull.

We can see a step of the algorithm in practice in figure 6. The orange points are points that have already been inserted into the convex hull in previous iterations of the algorithm. The last point to be inserted into the convex hull is point $(0.6, 0.4)$. We can find the bearing to all other points by finding the clockwise angle between the orange line that is parallel to the y axis and the line segments that lead from the last point to all the other points (drawn in blue). By looking at the diagram it is clear that the point with the smallest bearing is the point $(1, 0)$. This point will then be included in the convex hull and the algorithm will repeat.

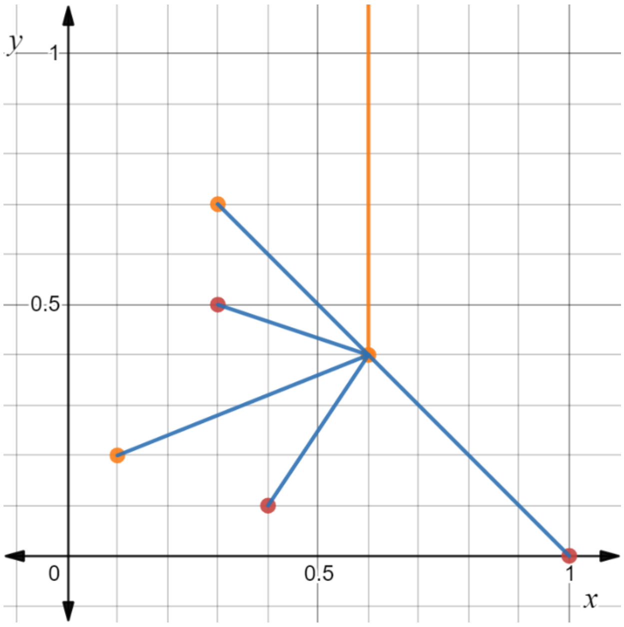

Figure 6: One step in making a convex hull (made by myself using desmos)

After this algorithm terminates, we will have a set containing all the points that are part of the convex hull. Therefore, we have found the convex hull.

Checking if a point is inside a convex hull:

We have now found a way to build a convex hull from a set of points. However, if we want to use a convex hull to solve the problem, we must also be able to check whether or not a point is inside a convex hull or not. This can be done using the method below:

First, we can first call the target point $T = (x_t, y_t)$. We will then create a dummy point $D = (\infty, y_t)$. We can then create a line segment that goes from point T to point D. This will be useful from the observation below.

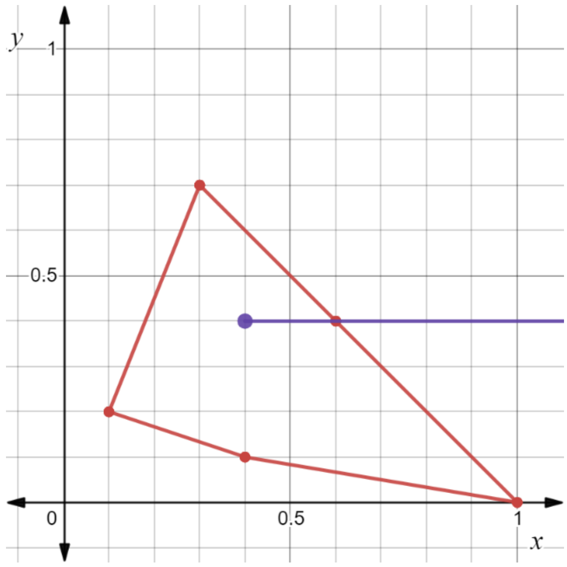

Figure 7: A point inside a convex hull (made by myself using desmos)

From figure 7, we can see that the line segment from the target point to the dummy point will intersect the convex hull 1 time. This will be the case as there will be 1 line segment to the left of the target point and 1 line segment to the right of the target point in the case that the point is inside the convex hull. Therefore, by checking that the number of times a line segment from the target point to a dummy point intersecting the convex hull is equal to one, we can check whether or not a point is inside a convex hull.

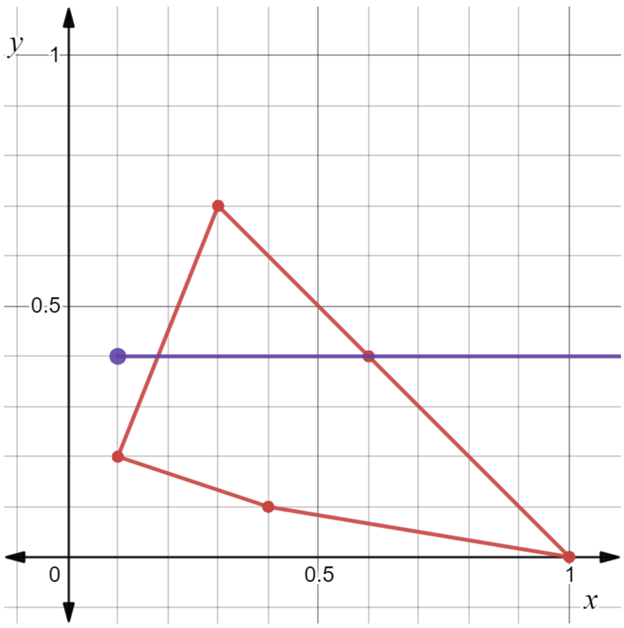

Figure 8: A point to the left of a convex hull (made by myself using desmos)

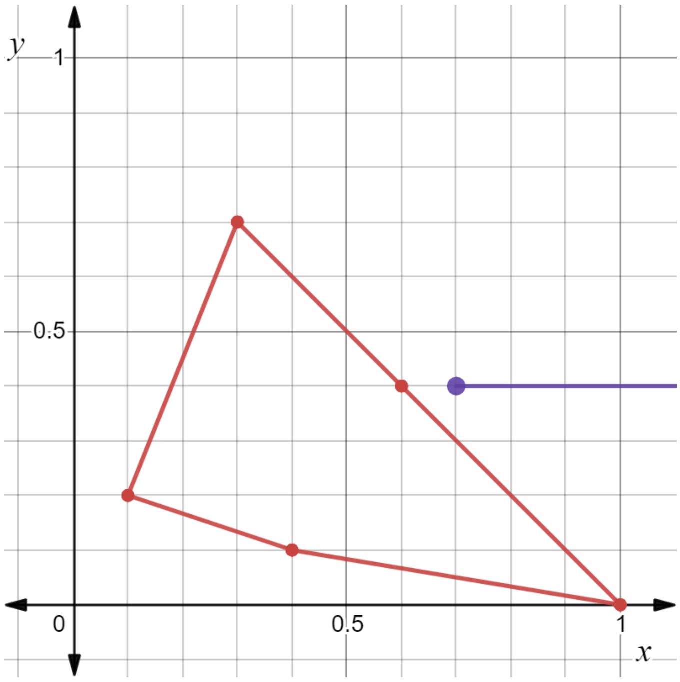

Figure 9: A point to the right of a convex hull (made by myself using desmos)

Looking at figures 8 and 9, it is clear that if a point is outside of a convex hull, the line segment between that point and the dummy point will intersect the convex hull either 0 or 2 times. Therefore, we can conclude that if a point is inside a convex hull, the line segment between the target point and the dummy point will intersect the convex hull exactly once.

Proving the triangulation of a point is inside a convex hull:

When finding the solution to the problem in 2 dimensions we asserted that if a target point is inside the convex hull, it can be made by combining 3 different pre-existing colors. However, no explanation or proof was given for this statement. In this section, this assertion will be proved using induction. It is worth noting that this proof is completely original and was invented by myself.

In this section we will assume that the set ch will be the set of points that are in the convex hull and $|ch|$ the number of points in the convex hull.

Lemma: Given that a point is inside a convex hull, there must be a triangle made using the vertices of the convex hull that encloses the point.

Step 1:

The base case of this proof will be when $|ch| = 3$ as we cannot make a polygon using less than 3 points. This case will match the lemma as a triangle will enclose the entire area of the triangle, hence, any point inside a triangle will be enclosed by the triangle.

Step 2:

We will assume the lemma to be true when $|ch| = k$ where $k \geq 3$ and where ch makes up a valid convex hull.

Step 3:

We will now show that the lemma is true when $|ch| = k + 1$ assuming that the lemma is true when $|ch| = k$. In order to prove this, we can see how the new point affects the existing convex hull.

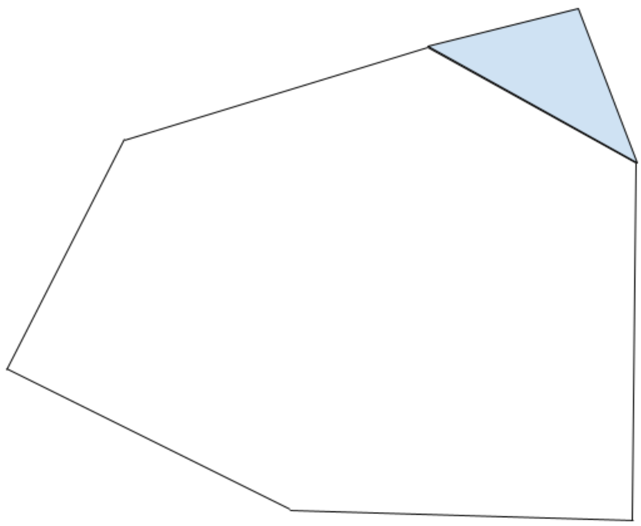

Figure 10: Adding a point to a convex hull (made by myself using google drawings)

In figure 10, the polygon excluding the blue triangle denotes the convex hull when $|ch| = k$ and the blue area is the newly added area by adding another point into the convex hull. By connecting the newly added point to the 2 points with minimum distance in the existing convex hull, we can create a new triangle that encompasses the area added to the convex hull (shown in blue). Therefore, assuming that the lemma holds true when $|ch| = k$ it also holds true when $|ch| = k + 1$.

Step 4:

If the lemma holds for $|ch| = k$, it holds for $|ch| = k + 1$. Since the lemma holds for $|ch| = 3$, by the Principle of Mathematical Induction, the lemma must hold for all ch such that $|ch| \geq 3$.

Real life applications:

We have now come up with a simple solution that is able to solve the problem. As stated in the introduction, we will now apply the solution that we found to 8 pre-existing colors and 3 target colors. The 8 pre-existing colors which represent the colors that I already have are shown in table 1.

Table 1: Pre-existing colors and coordinate mappings

Pre-existing colors and name of colors	Mapped pre-existing colors
$7:1:2$ (strong red)	$(\frac{7}{10}, \frac{1}{10})$
$1:7:2$ (strong khaki)	$(\frac{1}{10}, \frac{7}{10})$
$1:1:0$ (pure orange)	$(\frac{1}{2}, \frac{1}{2})$
$7:3:13$ (dark low saturated crimson)	$(\frac{7}{23}, \frac{3}{23})$
$2:7:1$ (pure khaki)	$(\frac{2}{10}, \frac{7}{10})$
$3:7:18$ (dark grayish bluish cyan)	$(\frac{3}{28}, \frac{7}{28})$
$30:19:1$ (pure orange red)	$(\frac{30}{50}, \frac{19}{50})$
$2:2:1$ (strong orange)	$(\frac{2}{5}, \frac{2}{5})$

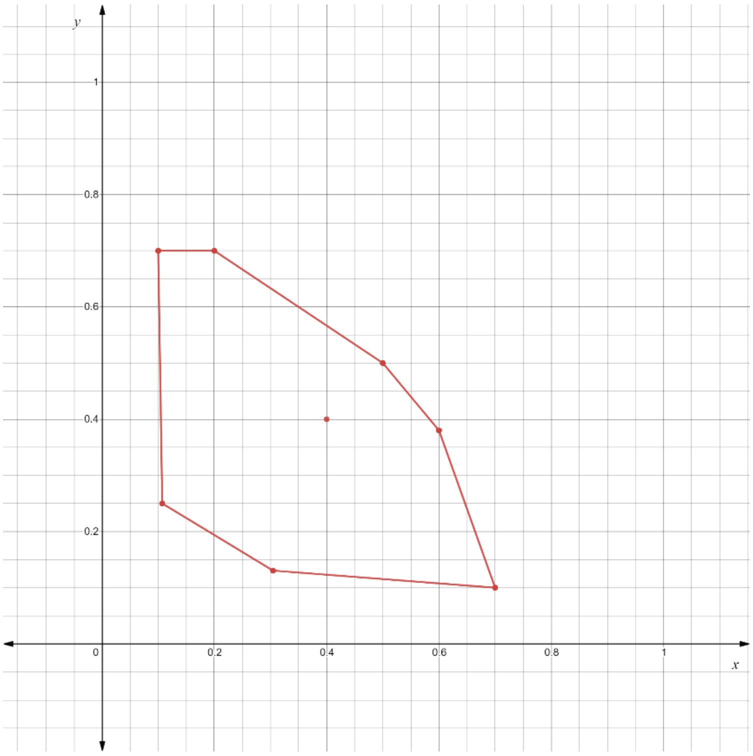

Figure 11: A convex hull made using the pre-existing points (made by myself using desmos)

Figure 11 shows the convex hull using the points shown in table 1. It is clear from the diagram that there are no interior angles that exceed $180°$, making the polygon a convex hull. We will now try answering whether or not some target colors can be made using the pre-existing colors below.

Query 1: $7:12:1$ (pure orange) or point $(\frac{7}{20}, \frac{12}{20})$

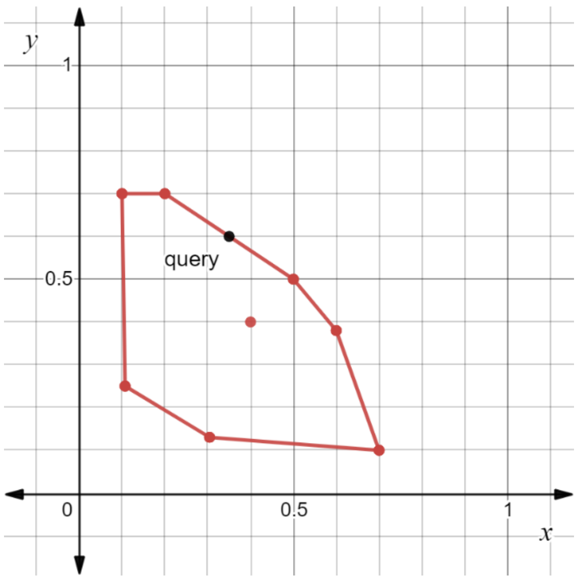

Figure 12: The final convex hull and the first query (made by myself using desmos)

From figure 12, we can clearly see that the point $(\frac{7}{20}, \frac{12}{20})$ lies on the line segment between point $(\frac{7}{10}, \frac{2}{10})$ and $(\frac{1}{2}, \frac{1}{2})$. Therefore, we can conclude that it is possible to make the ratio $7{:}\,12{:}\,1$ and it would take 2 different pre-existing colors to make this target color.

Query 2: $3{:}\,5{:}\,2$ (dark strong orange) or point $(\frac{3}{10}, \frac{5}{10})$

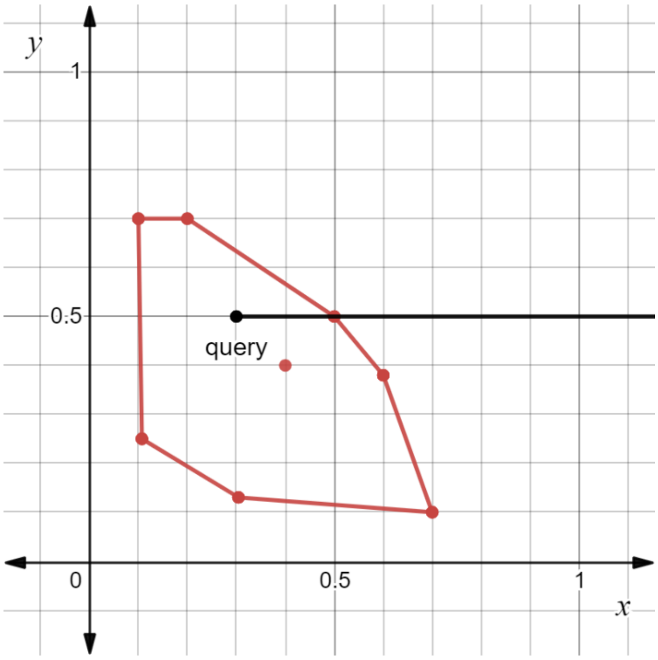

Figure 13: The final convex hull and the second query (made by myself using desmos)

From figure 13, we can see that line segment between point $(\frac{3}{10}, \frac{5}{10})$ and point $(\infty, \frac{5}{10})$ will intersect the convex hull once at point $(\frac{1}{2}, \frac{1}{2})$. Therefore, we can conclude that the target point is inside the convex hull and we will have to mix 3 different pre-existing colors to make the target color.

Query 3: $0:5:5$ (dark medium olive) or point $(\frac{0}{10}, \frac{5}{10})$

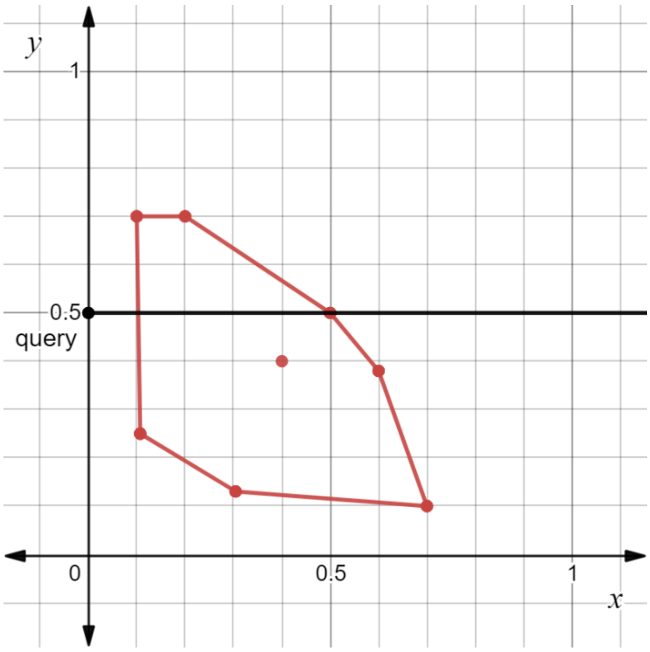

Figure 14: The final convex hull and the third query (made by myself using desmos)

From figure 14, we can see that line segment between point $(\frac{0}{10}, \frac{5}{10})$ and point $(\infty, \frac{5}{10})$ will intersect the convex hull twice at approximately point $(\frac{1}{10}, \frac{5}{10})$ and at point $(\frac{1}{2}, \frac{1}{2})$. Therefore, we can conclude that the target point is outside the convex hull and it will be impossible to make this target color using the pre-existing colors that we have.

Conclusion and evaluation:

By investigating the problem, we have come up with 2 solutions, one complex solution that uses $3D$ geometry and one relatively simpler solution that uses $2D$ geometry. This means that we have successfully met the goal of the investigation which was to create a solution to the problem listed in the introduction. In particular we made a convex hull containing the 8 pre-existing points that we had and afterwards, answered 3 queries based on target colors that we wanted to make.

I found this investigation to be fascinating because we were able to use seemingly unrelated areas of mathematics to solve the problem. When looking at the problem in the beginning it seems to be a combinatorial optimization problem (problems in which a certain number of elements from a set of elements has to be chosen) as we have to choose a certain number of pre-existing colors to mix in order to make a new target color. However, this turned out to be false as we could use 3D vector geometry to find whether or not a target point can be made. This solution was unfortunately quite slow when dealing with many pre-existing colors. We then looked at optimizing this solution into a 2D geometry problem where we brought in the idea of convex hulls to more efficiently check whether or not a target point can be made using 3 pre-existing colors.

One way that we could extend this investigation is by instead of using red, yellow, and blue we could also try solving the problem for cyan, magenta, yellow and black which are the colors used in printers. This would have more practical uses as the problem that we discussed could have a large impact in printers, as printers have to mix colors in different ratios to print on paper. Since, printers use the CMYK color scheme instead of a more traditional RYB color scheme, making a solution that uses 4 colors instead of 3 would have more real-life applications.

Another way that we could extend this investigation is by instead of just finding the minimum number of pre-existing colors that we have to mix, we could also find the ratios in which those pre-existing colors are mixed. This will result in more practical applications as by having this information, one could actually mix the colors instead of just knowing how many colors they have to mix to make their target color.

Works Cited:

"BOI 2020 Online Mirror - Day 1 Tutorial." *Codeforces*, codeforces.com/blog/entry/80490.